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(54) **SOURCE BIAS TEMPERATURE
COMPENSATION FOR READ AND
PROGRAM VERIFY OPERATIONS ON A
MEMORY DEVICE**

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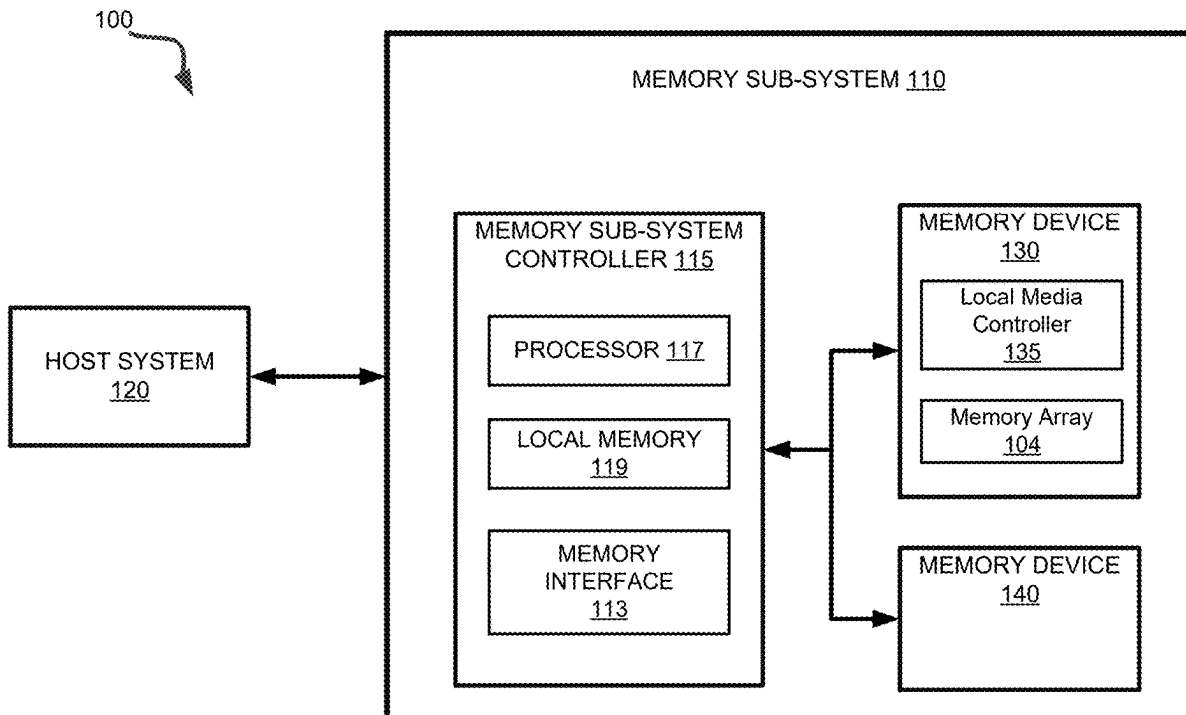
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(57) **ABSTRACT**

Control logic in a memory device receives a request to perform a memory access operation on a memory array of the memory device and determines an operating temperature of the memory device. The control logic further modifies a default magnitude of a source voltage signal based on the operating temperature to form a modified source voltage signal, causes the modified source voltage signal to be applied to the memory array, and performs the memory access operation on the memory array.



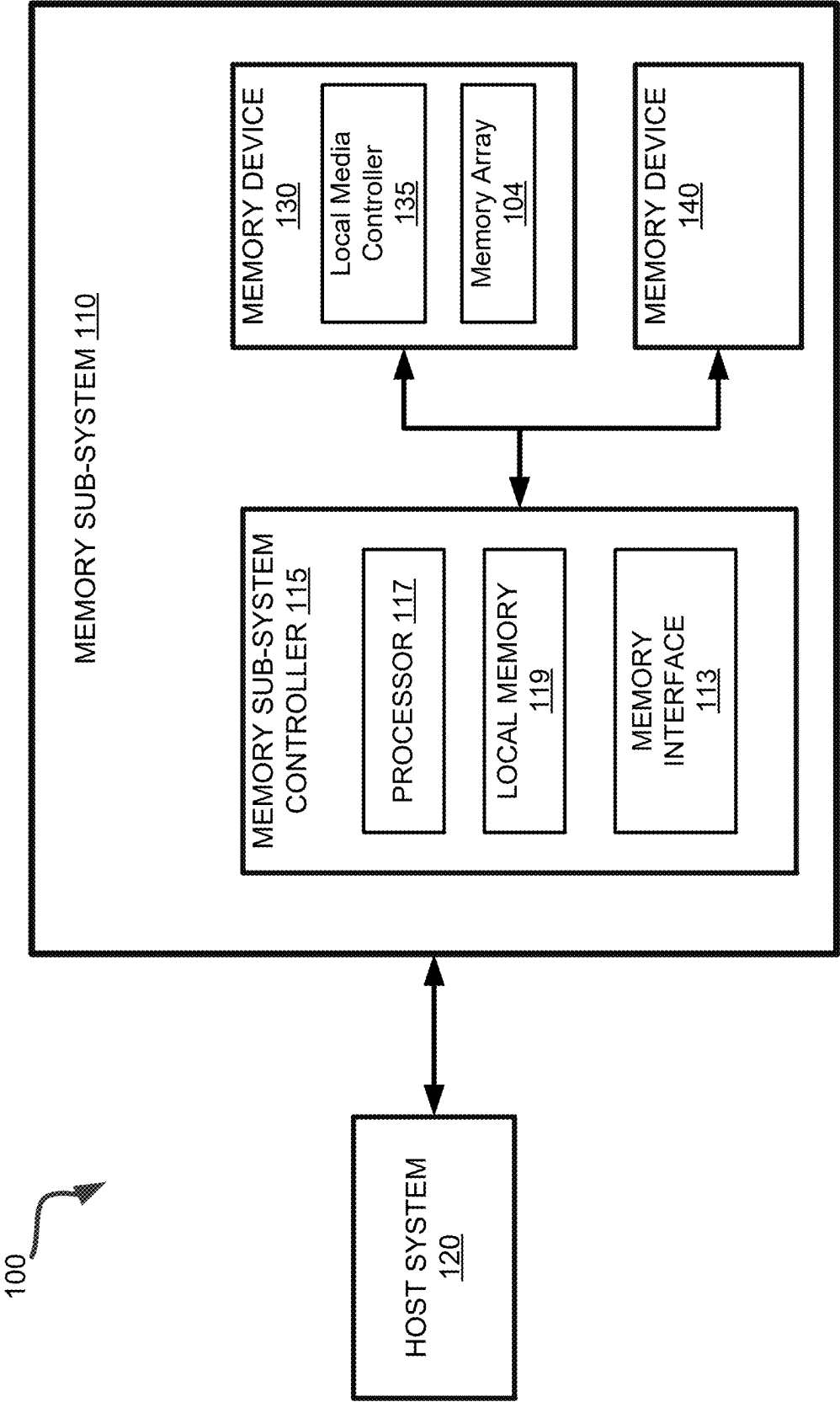


FIG. 1A

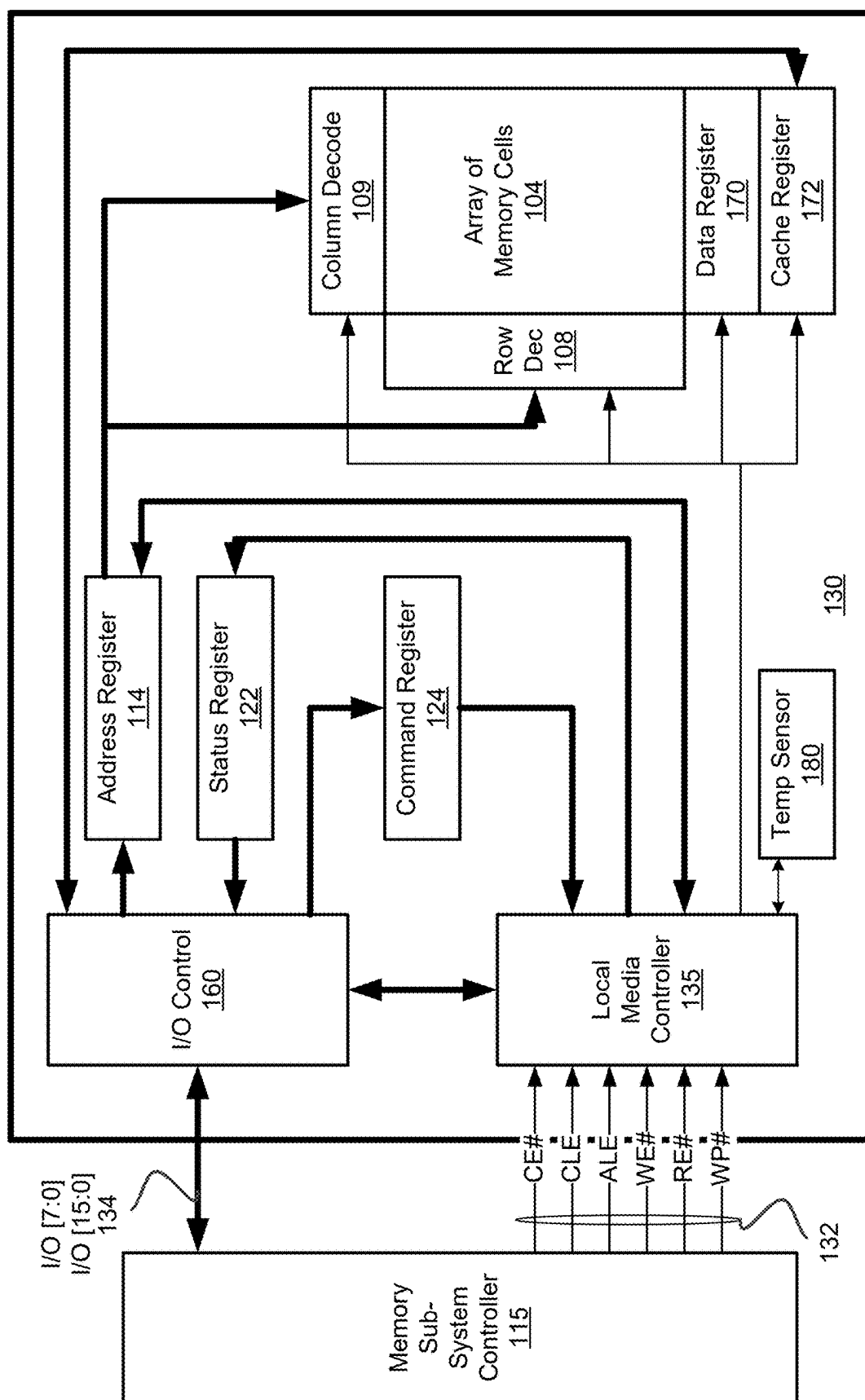


FIG. 1B

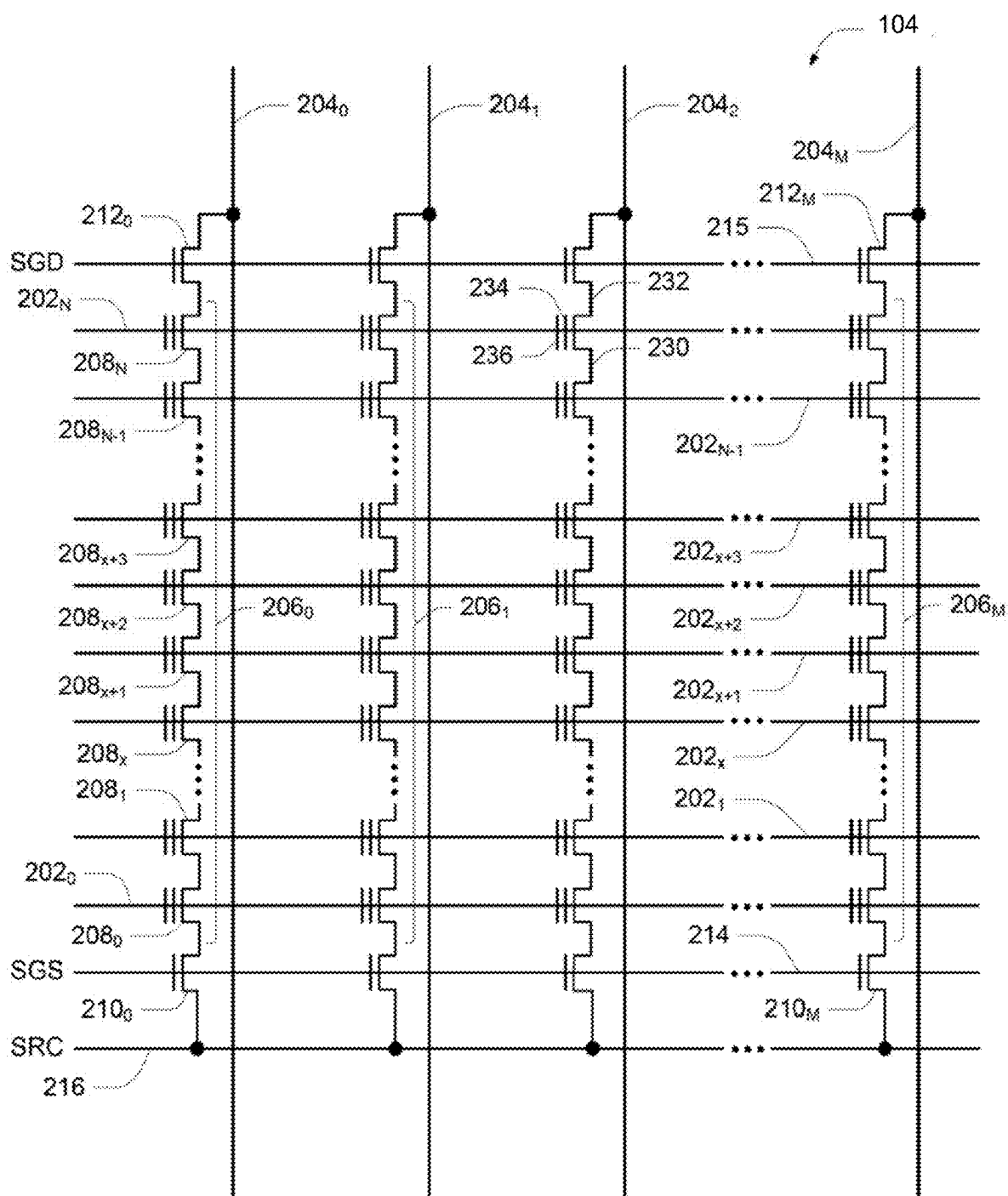
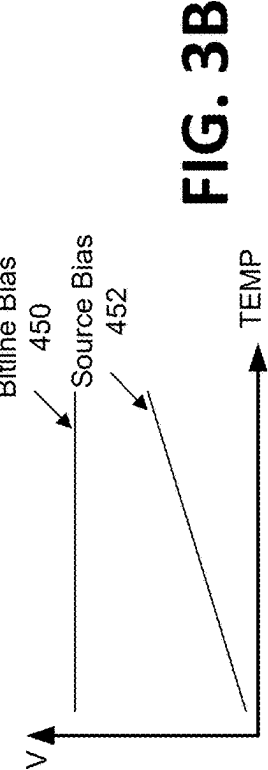
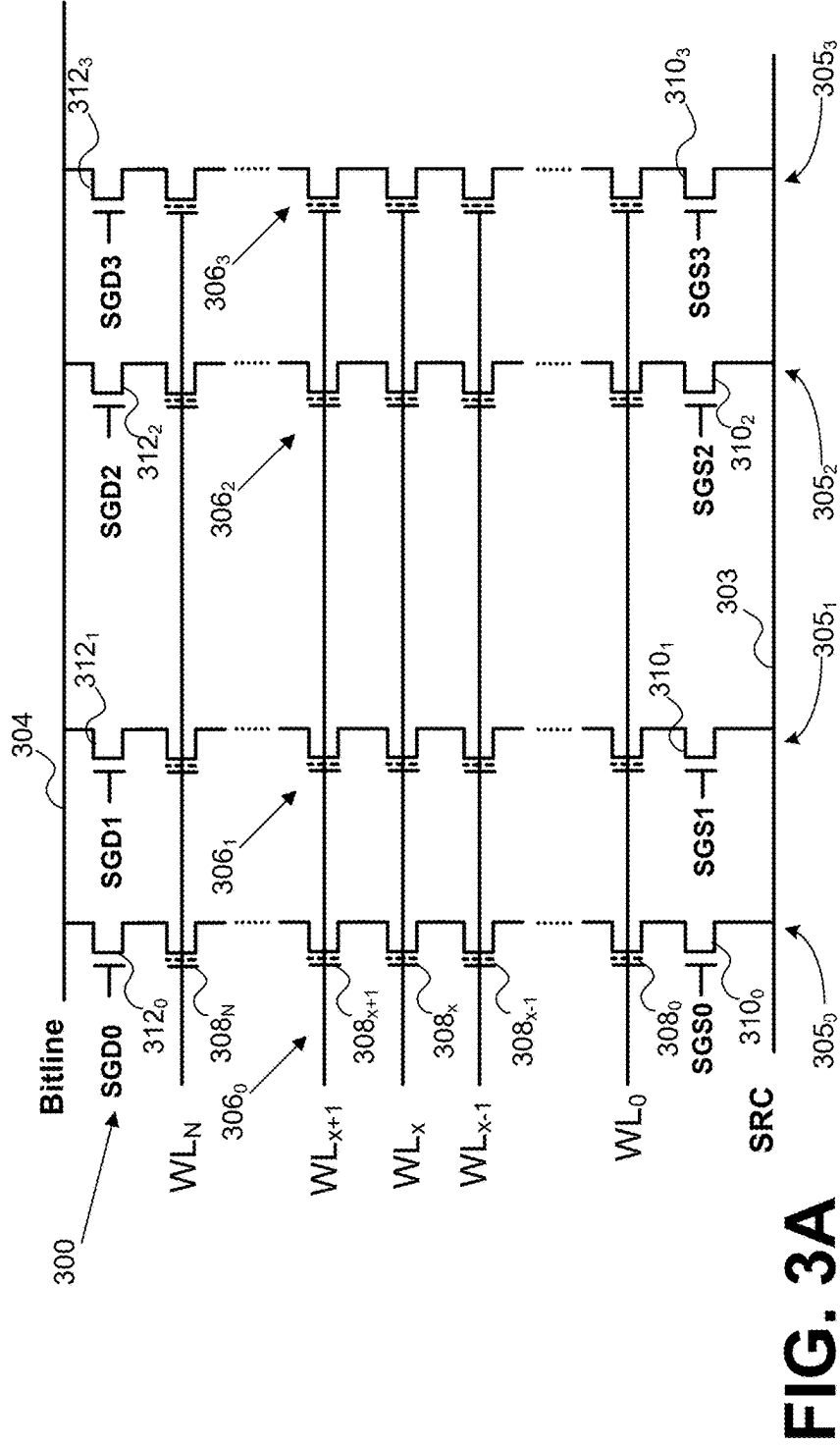
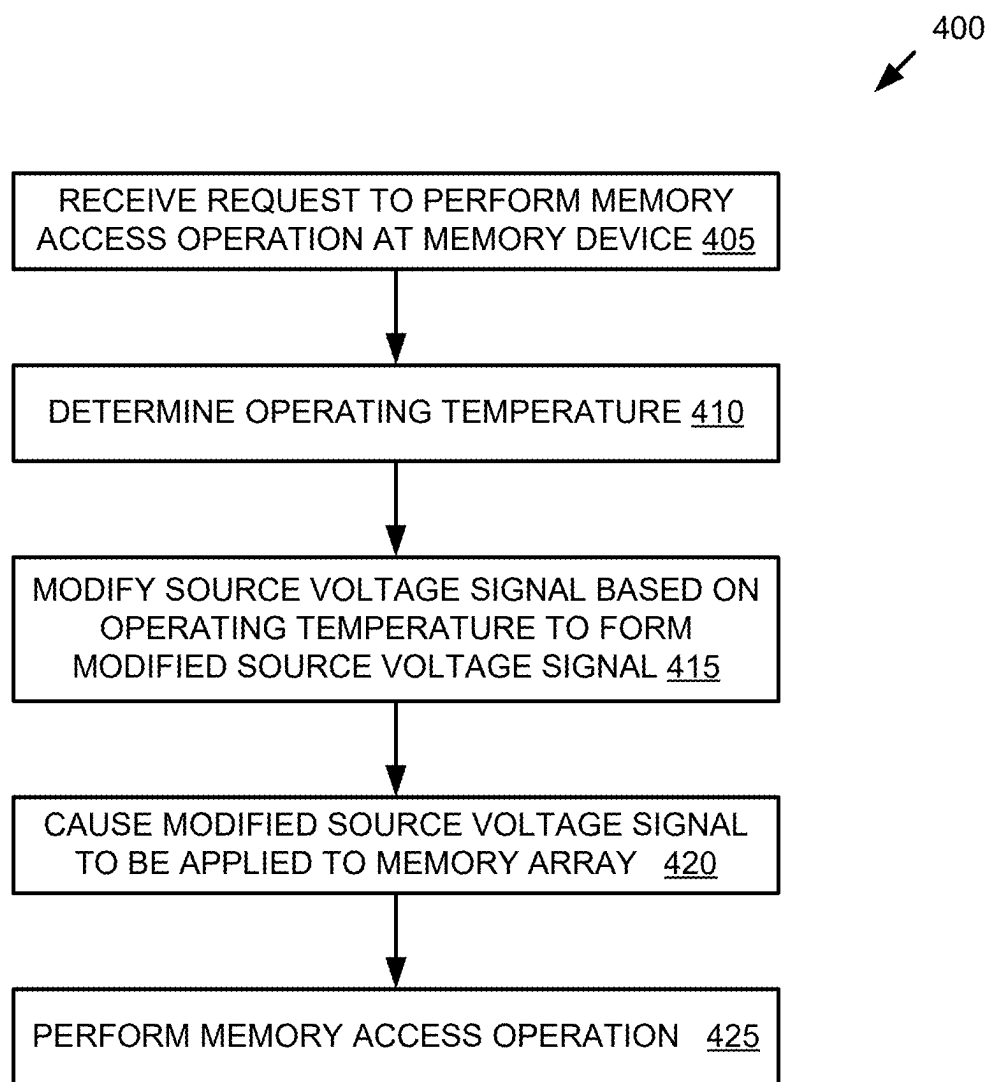


FIG. 2



**FIG. 4**

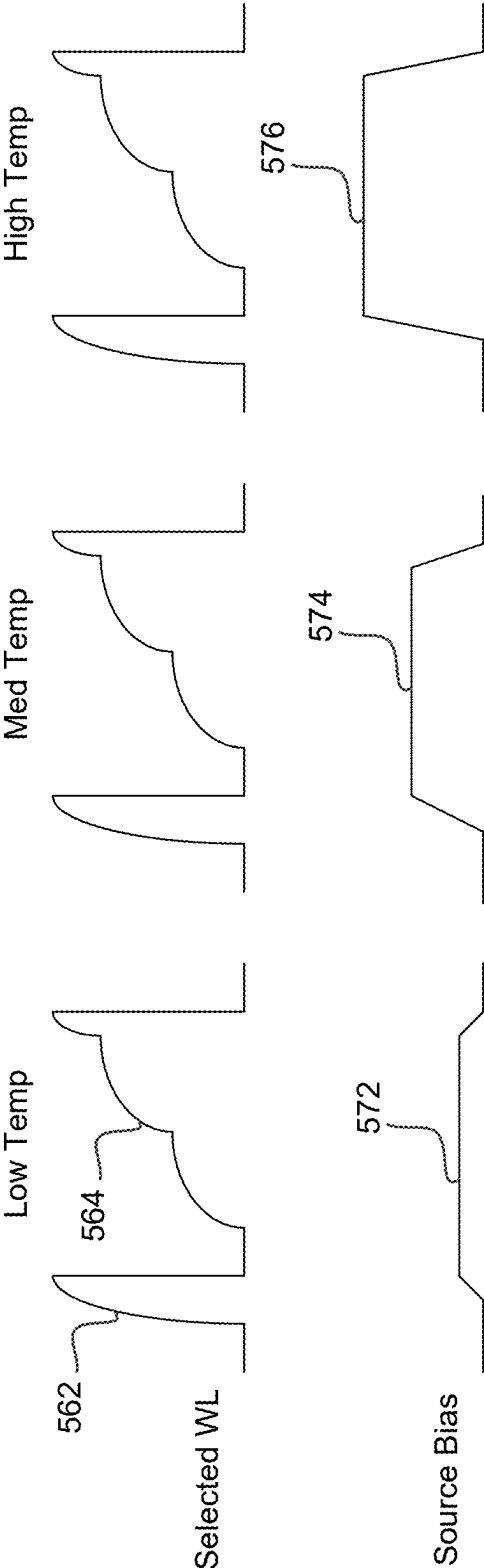


FIG. 5

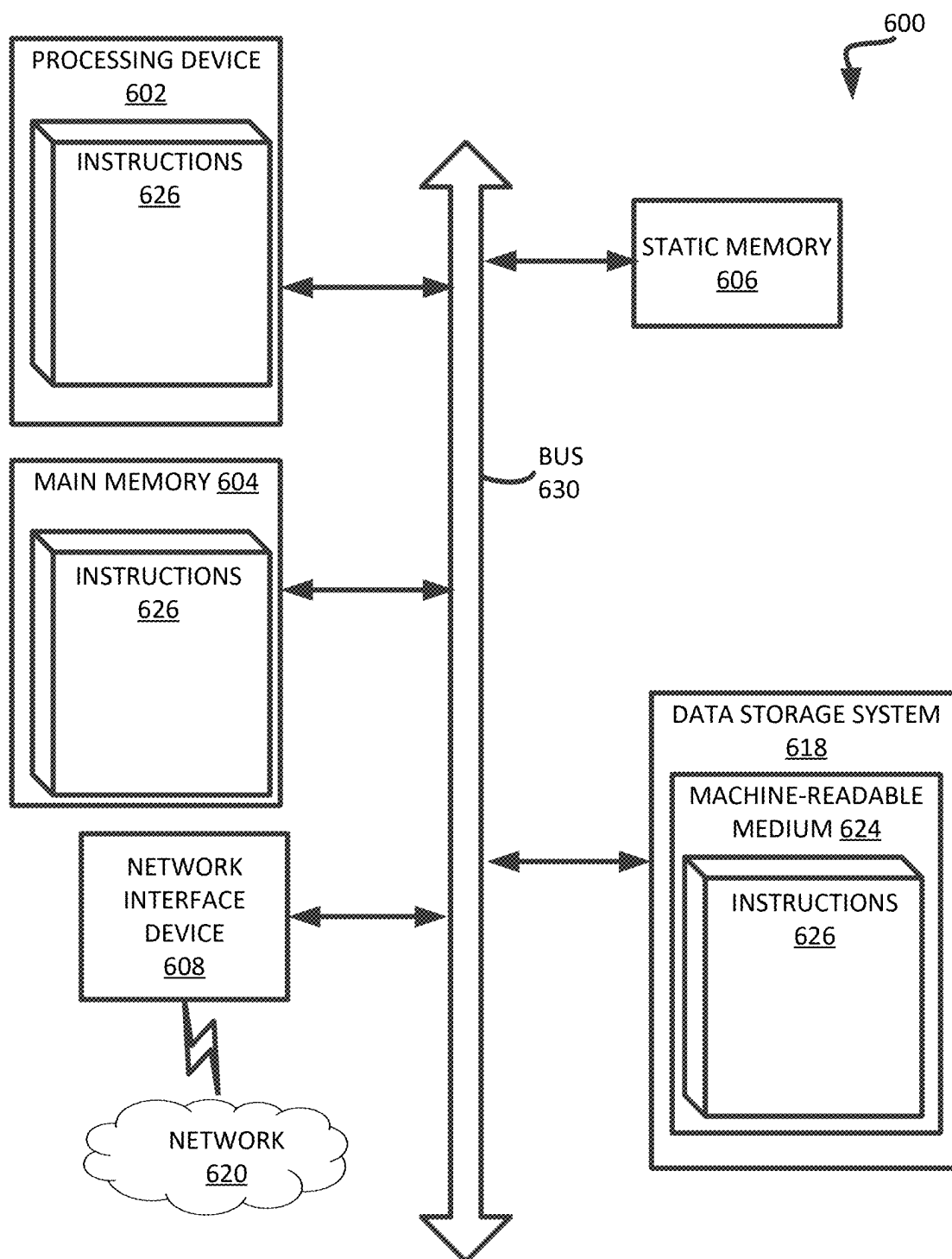


FIG. 6

SOURCE BIAS TEMPERATURE COMPENSATION FOR READ AND PROGRAM VERIFY OPERATIONS ON A MEMORY DEVICE

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/994,907, filed Nov. 28, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/292,201, filed Dec. 21, 2021, the entire contents of each of which are hereby incorporated by reference herein.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate generally to memory sub-systems, and more specifically, relate to source bias temperature compensation for read and program verify operations on a memory device of a memory sub-system.

BACKGROUND

[0003] A memory sub-system can include one or more memory devices that store data. The memory devices can be, for example, non-volatile memory devices and volatile memory devices. In general, a host system can utilize a memory sub-system to store data at the memory devices and to retrieve data from the memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure.

[0005] FIG. 1A illustrates an example computing system that includes a memory sub-system in accordance with some embodiments of the present disclosure.

[0006] FIG. 1B is a block diagram of a memory device in communication with a memory sub-system controller of a memory sub-system, in accordance with some embodiments of the present disclosure.

[0007] FIG. 2 is a schematic of portions of an array of memory cells as could be used in a memory of the type described with reference to FIG. 1B in accordance with some embodiments of the present disclosure.

[0008] FIG. 3A is a schematic of portions of an array of memory cells implementing source bias temperature compensation for read and program verify operations in accordance with some embodiments of the present disclosure.

[0009] FIG. 3B is a graph illustrating source bias temperature compensation for read and program verify operations on a memory device in accordance with some embodiments of the present disclosure.

[0010] FIG. 4 is a flow diagram of an example method of source bias temperature compensation for read and program verify operations on a memory device of a memory sub-system in accordance with some embodiments of the present disclosure.

[0011] FIG. 5 is a diagram illustrating example waveforms for source bias temperature compensation for read and program verify operations on a memory device in accordance with some embodiments of the present disclosure.

[0012] FIG. 6 is a block diagram of an example computer system in which embodiments of the present disclosure can operate.

DETAILED DESCRIPTION

[0013] Aspects of the present disclosure are directed to source bias temperature compensation for read and program verify operations on a memory device of a memory sub-system. A memory sub-system can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of storage devices and memory modules are described below in conjunction with FIG. 1. In general, a host system can utilize a memory sub-system that includes one or more components, such as memory devices that store data. The host system can provide data to be stored at the memory sub-system and can request data to be retrieved from the memory sub-system.

[0014] A memory sub-system can include high density non-volatile memory devices where retention of data is desired when no power is supplied to the memory device. For example, NAND memory, such as 3D flash NAND memory, offers storage in the form of compact, high density configurations. A non-volatile memory device is a package of one or more dice, each including one or more planes. For some types of non-volatile memory devices (e.g., NAND memory), each plane includes a set of physical blocks. Each block includes a set of pages. Each page includes a set of memory cells ("cells"). A cell is an electronic circuit that stores information. Depending on the cell type, a cell can store one or more bits of binary information, and has various logic states that correlate to the number of bits being stored. The logic states can be represented by binary values, such as "0" and "1", or combinations of such values.

[0015] A memory device can be made up of bits arranged in a two-dimensional or a three-dimensional grid. Memory cells are etched onto a silicon wafer in an array of columns (also hereinafter referred to as bitlines) and rows (also hereinafter referred to as wordlines). A wordline can refer to one or more rows of memory cells of a memory device that are used with one or more bitlines to generate the address of each of the memory cells. The intersection of a bitline and wordline constitutes the address of the memory cell. A block hereinafter refers to a unit of the memory device used to store data and can include a group of memory cells, a wordline group, a wordline, or individual memory cells. One or more blocks can be grouped together to form separate partitions (e.g., planes) of the memory device in order to allow concurrent operations to take place on each plane.

[0016] Bit flip errors can occur in certain memory devices when there is not enough separation between respective threshold voltages (V_t) of two adjacent bit levels (also referred to as "states"). Typically, each binary value stored in a memory cell has a different associated threshold voltage, with the lowest binary value having the highest threshold voltage, the highest binary value having the lowest threshold voltage, and intermediate states having progressively different threshold voltage values. For example, a memory cell configured as triple level cell (TLC) memory can have eight states, with each state having a corresponding V_t . Similarly, a memory cell configured as quad level cell (QLC) memory can have 16 states, with each state having a corresponding V_t . In certain memory devices, bit flip errors can be reduced (e.g., minimized) through providing a better separation of levels in a threshold voltage (V_t) distribution. The separation between two adjacent levels is reduced, however, when more bits are stored per memory cell.

[0017] In many memory devices, the level separation in threshold voltages becomes further reduced (or shifted) due

to changes in environmental conditions, such as cross-temperature effects. Cross temperature negatively impacts level separation in situations where the memory cell operates (e.g., is read) at a temperature range which is different from a temperature at which the memory cell was programmed. For example, cross temperature effects can arise when data is read from a memory cell at a temperature that is different from the temperature at which data was written into the memory cell. Cross-temperature-induced errors can be accumulated by one or both of shifted levels that cross thresholds boundaries and/or overlapping levels causing increased number of bit flip errors. Bit flip errors reduce reliability and data retention capability as a result of the increased error rate. As the difference between a data write temperature and a data read temperature increases, so does the error rate of the data as a result of level shift and level overlap.

[0018] Certain memory devices and memory sub-systems attempt to reduce the error rates using a variety of techniques, including adjusting the magnitude of certain voltage signals applied to the memory array during different operations. This can include determining a compensation offset value for the voltage signals to account for a given memory cell's shift in threshold voltage or the widening of a voltage distribution. Since the threshold voltage shift can vary depending on process variations in each memory cell, the location of the memory cell (i.e., die to die variations), and the number of program/erase cycles performed on the cell, such a calibration process can be complicated. For example, certain memory devices attempt to adjust the magnitude of a bitline voltage signal applied during a read or a program verify operation being performed on specific cells of the memory array. This adjustment can take the form of a temperature compensation value used to modify the voltage magnitude based on a temperature measured at the start of the read or program verify operation. Adjusting the bitline voltage signal in this manner can account for threshold voltage shift, but also introduces other complications. For example, in a multi-plane memory device where asynchronous operations are performed concurrently (e.g., at least partially overlapping in time) on separate memory planes, the operations can start and end at different times. Accordingly, measuring the temperature at the start of one operation on a first plane, and adjusting the magnitude of the bitline voltage signal based on the temperature measurement, can introduce a level of noise on the bitline which can negatively impact a potentially sensitive an unfinished operation occurring concurrently on a second plane of the memory device. In addition, even despite any noise introduced, temperature compensation for the bitline voltage signal still only results in limited improvements to the margins between levels in a threshold voltage (V_t) distribution of the memory device. In particular, the space between levels (e.g., the read window budget (RWB)) may not be adequate to prevent errors during the read or program verify operation, particularly in memory devices configured to store multiple bits per cell.

[0019] Aspects of the present disclosure address the above and other deficiencies by providing source bias temperature compensation for read and program verify operations on a memory device of a memory sub-system. In one embodiment, control logic on the memory device can adjust a source voltage signal, rather than the bitline voltage signal, based on a temperature measured at the start of a read or a program verify operation. For example, upon receiving a

request to perform the read or program verify operation, the control can measure a temperature of the memory die (e.g., using a built-in temperature sensor or requesting a temperature reading from an external sensor) and determine a temperature compensation value for a source voltage signal, such as a source voltage offset (e.g., by performing a calculation using derived formulas or by using a look-up table or other data structure stored on the memory device). Using the determined temperature compensation value, the control logic can modify the source voltage signal before causing the source voltage signal to be applied to a source line of the memory array, and can then proceed with performing the read or program verify operation, such as by causing a read or program verify signal to be applied to a wordline associated with the selected memory cell or cells. In general, the temperature compensation value increases, thereby causing the modified source voltage signal to increase, as the temperature at which the read or program verify operation is performed increases. In one embodiment, the bitline voltage signal remains constant (i.e., unmodified) during the read or program verify operation.

[0020] Advantages of this approach include, but are not limited to, improved performance in the memory device. Modifying the source voltage signal using a temperature compensation value based on temperature counteracts threshold voltage shift in the memory device and reduces the occurrence of errors during read and program verify operations. The techniques described herein increase the drain-to-source voltage differential (V_{ds}), which, in turn, causes an increased string current through the selected memory cell(s). This offsets any current degradation present at low operating temperatures and stabilizes the string current throughout the operating temperature range, which limits threshold voltage distribution widening and allows the read and program verify operations to be performed with a larger read window budget. In addition, compensating for temperature at the source voltage signal reduces cell-to-cell interference and prevents the need to increase the pass voltage, thereby preventing read disturb errors.

[0021] FIG. 1A illustrates an example computing system 100 that includes a memory sub-system 110 in accordance with some embodiments of the present disclosure. The memory sub-system 110 can include media, such as one or more volatile memory devices (e.g., memory device 140), one or more non-volatile memory devices (e.g., memory device 130), or a combination of such.

[0022] A memory sub-system 110 can be a storage device, a memory module, or a hybrid of a storage device and memory module. Examples of a storage device include a solid-state drive (SSD), a flash drive, a universal serial bus (USB) flash drive, an embedded Multi-Media Controller (eMMC) drive, a Universal Flash Storage (UFS) drive, a secure digital (SD) card, and a hard disk drive (HDD). Examples of memory modules include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), and various types of non-volatile dual in-line memory modules (NVDIMMs).

[0023] The computing system 100 can be a computing device such as a desktop computer, laptop computer, network server, mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), Internet of Things (IoT) enabled device, embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked

commercial device), or such computing device that includes memory and a processing device.

[0024] The computing system **100** can include a host system **120** that is coupled to one or more memory sub-systems **110**. In some embodiments, the host system **120** is coupled to different types of memory sub-system **110**. FIG. 1A illustrates one example of a host system **120** coupled to one memory sub-system **110**. As used herein, “coupled to” or “coupled with” generally refers to a connection between components, which can be an indirect communicative connection or direct communicative connection (e.g., without intervening components), whether wired or wireless, including connections such as electrical, optical, magnetic, etc.

[0025] The host system **120** can include a processor chip-set and a software stack executed by the processor chip-set. The processor chip-set can include one or more cores, one or more caches, a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., PCIe controller, SATA controller). The host system **120** uses the memory sub-system **110**, for example, to write data to the memory sub-system **110** and read data from the memory sub-system **110**.

[0026] The host system **120** can be coupled to the memory sub-system **110** via a physical host interface. Examples of a physical host interface include, but are not limited to, a serial advanced technology attachment (SATA) interface, a peripheral component interconnect express (PCIe) interface, universal serial bus (USB) interface, Fibre Channel, Serial Attached SCSI (SAS), a double data rate (DDR) memory bus, Small Computer System Interface (SCSI), a dual in-line memory module (DIMM) interface (e.g., DIMM socket interface that supports Double Data Rate (DDR)), etc. The physical host interface can be used to transmit data between the host system **120** and the memory sub-system **110**. The host system **120** can further utilize an NVM Express (NVMe) interface to access the memory components (e.g., memory devices **130**) when the memory sub-system **110** is coupled with the host system **120** by the PCIe interface. The physical host interface can provide an interface for passing control, address, data, and other signals between the memory sub-system **110** and the host system **120**. FIG. 1A illustrates a memory sub-system **110** as an example. In general, the host system **120** can access multiple memory sub-systems via a same communication connection, multiple separate communication connections, and/or a combination of communication connections.

[0027] The memory devices **130**, **140** can include any combination of the different types of non-volatile memory devices and/or volatile memory devices. The volatile memory devices (e.g., memory device **140**) can be, but are not limited to, random access memory (RAM), such as dynamic random access memory (DRAM) and synchronous dynamic random access memory (SDRAM).

[0028] Some examples of non-volatile memory devices (e.g., memory device **130**) include negative-and (NAND) type flash memory and write-in-place memory, such as three-dimensional cross-point (“3D cross-point”) memory. A cross-point array of non-volatile memory can perform bit storage based on a change of bulk resistance, in conjunction with a stackable cross-gridded data access array. Additionally, in contrast to many flash-based memories, cross-point non-volatile memory can perform a write in-place operation, where a non-volatile memory cell can be programmed without the non-volatile memory cell being previously

erased. NAND type flash memory includes, for example, two-dimensional NAND (2D NAND) and three-dimensional NAND (3D NAND).

[0029] Each of the memory devices **130** can include one or more arrays of memory cells. One type of memory cell, for example, single level cells (SLC) can store one bit per cell. Other types of memory cells, such as multi-level cells (MLCs), triple level cells (TLCs), and quad-level cells (QLCs), can store multiple bits per cell. In some embodiments, each of the memory devices **130** can include one or more arrays of memory cells such as SLCs, MLCs, TLCs, QLCs, or any combination of such. In some embodiments, a particular memory device can include an SLC portion, and an MLC portion, a TLC portion, or a QLC portion of memory cells. The memory cells of the memory devices **130** can be grouped as pages that can refer to a logical unit of the memory device used to store data. With some types of memory (e.g., NAND), pages can be grouped to form blocks.

[0030] Although non-volatile memory components such as a 3D cross-point array of non-volatile memory cells and NAND type flash memory (e.g., 2D NAND, 3D NAND) are described, the memory device **130** can be based on any other type of non-volatile memory, such as read-only memory (ROM), phase change memory (PCM), self-selecting memory, other chalcogenide based memories, ferroelectric transistor random-access memory (FeTRAM), ferroelectric random access memory (FeRAM), magneto random access memory (MRAM), Spin Transfer Torque (STT)-MRAM, conductive bridging RAM (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), negative-or (NOR) flash memory, electrically erasable programmable read-only memory (EEPROM).

[0031] A memory sub-system controller **115** (or controller **115** for simplicity) can communicate with the memory devices **130** to perform operations such as reading data, writing data, or erasing data at the memory devices **130** and other such operations. The memory sub-system controller **115** can include hardware such as one or more integrated circuits and/or discrete components, a buffer memory, or a combination thereof. The hardware can include a digital circuitry with dedicated (i.e., hard-coded) logic to perform the operations described herein. The memory sub-system controller **115** can be a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.), or other suitable processor.

[0032] The memory sub-system controller **115** can include a processor **117** (e.g., a processing device) configured to execute instructions stored in a local memory **119**. In the illustrated example, the local memory **119** of the memory sub-system controller **115** includes an embedded memory configured to store instructions for performing various processes, operations, logic flows, and routines that control operation of the memory sub-system **110**, including handling communications between the memory sub-system **110** and the host system **120**.

[0033] In some embodiments, the local memory **119** can include memory registers storing memory pointers, fetched data, etc. The local memory **119** can also include read-only memory (ROM) for storing micro-code. While the example memory sub-system **110** in FIG. 1A has been illustrated as including the memory sub-system controller **115**, in another embodiment of the present disclosure, a memory sub-system

110 does not include a memory sub-system controller **115**, and can instead rely upon external control (e.g., provided by an external host, or by a processor or controller separate from the memory sub-system).

[0034] In general, the memory sub-system controller **115** can receive commands or operations from the host system **120** and can convert the commands or operations into instructions or appropriate commands to achieve the desired access to the memory devices **130**. The memory sub-system controller **115** can be responsible for other operations such as wear leveling operations, garbage collection operations, error detection and error-correcting code (ECC) operations, encryption operations, caching operations, and address translations between a logical address (e.g., logical block address (LBA), namespace) and a physical address (e.g., physical block address) that are associated with the memory devices **130**. The memory sub-system controller **115** can further include host interface circuitry to communicate with the host system **120** via the physical host interface. The host interface circuitry can convert the commands received from the host system into command instructions to access the memory devices **130** as well as convert responses associated with the memory devices **130** into information for the host system **120**.

[0035] The memory sub-system **110** can also include additional circuitry or components that are not illustrated. In some embodiments, the memory sub-system **110** can include a cache or buffer (e.g., DRAM) and address circuitry (e.g., a row decoder and a column decoder) that can receive an address from the memory sub-system controller **115** and decode the address to access the memory devices **130**.

[0036] In some embodiments, the memory devices **130** include local media controllers **135** that operate in conjunction with memory sub-system controller **115** to execute operations on one or more memory cells of the memory devices **130**. An external controller (e.g., memory sub-system controller **115**) can externally manage the memory device **130** (e.g., perform media management operations on the memory device **130**). In some embodiments, a memory device **130** is a managed memory device, which is a raw memory device **130** having control logic (e.g., local controller **135**) on the die and a controller (e.g., memory sub-system controller **115**) for media management within the same memory device package. An example of a managed memory device is a managed NAND (MNAND) device. Memory device **130**, for example, can represent a single die having some control logic (e.g., local media controller **135**) embodied thereon. In some embodiments, one or more components of memory sub-system **110** can be omitted.

[0037] In one embodiment, memory sub-system **110** includes a memory interface component **113**. Memory interface component **113** is responsible for handling interactions of memory sub-system controller **115** with the memory devices of memory sub-system **110**, such as memory device **130**. For example, memory interface component **113** can send memory access commands corresponding to requests received from host system **120** to memory device **130**, such as program commands, read commands, or other commands (e.g., program verify commands). In addition, memory interface component **113** can receive data from memory device **130**, such as data retrieved in response to a read command or a confirmation that a program command was successfully performed. In some embodiments, the memory sub-system controller **115** includes at least a portion of the

memory interface **113**. For example, the memory sub-system controller **115** can include a processor **117** (e.g., a processing device) configured to execute instructions stored in local memory **119** for performing the operations described herein. In some embodiments, the memory interface component **113** is part of the host system **110**, an application, or an operating system.

[0038] In one embodiment, memory device **130** includes local media controller **135** and a memory array **104**. In one embodiment, control logic of local media controller **135** is configured to perform source bias temperature compensation for read and program verify operations being performed on memory array **104**. As described in more detail below, memory array **104** can include a number of strings of memory cells coupled between a common source (e.g., a source line or source plate) and a drain (e.g., a bitline or bitline portion). Each memory cell in the strings can be further coupled to a respective wordline. During memory access operations performed in response to a request (e.g., received from memory sub-system controller **115** or host system **120**), local media controller **135** can cause various voltage bias signals to be applied to memory array **104**. In one embodiment, local media controller **135** can adjust or modify the magnitudes of these signals based on temperature during performance of certain types of memory access operations, such as read operations or program verify operations, for example. In one embodiment, local media controller **135** receives a request to perform a memory access operation on memory array **104** of memory device **130** and determines an operating temperature of the memory device **130**. Local media controller **135** further modifies a default magnitude of a source voltage signal based on the operating temperature to form a modified source voltage signal, causes the modified source voltage signal to be applied to the memory array **104**, and performs the memory access operation on the memory array **104**, such as by causing a corresponding voltage signal to be applied to a selected wordline. Further details with regards to the operations of local media controller **135** are described below.

[0039] FIG. 1B is a simplified block diagram of a first apparatus, in the form of a memory device **130**, in communication with a second apparatus, in the form of a memory sub-system controller **115** of a memory sub-system (e.g., memory sub-system **110** of FIG. 1A), according to an embodiment. Some examples of electronic systems include personal computers, personal digital assistants (PDAs), digital cameras, digital media players, digital recorders, games, appliances, vehicles, wireless devices, mobile telephones and the like. The memory sub-system controller **115** (e.g., a controller external to the memory device **130**), may be a memory controller or other external host device.

[0040] Memory device **130** includes an array of memory cells **104** logically arranged in rows and columns. Memory cells of a logical row are typically connected to the same access line (e.g., a wordline) while memory cells of a logical column are typically selectively connected to the same data line (e.g., a bit line). A single access line may be associated with more than one logical row of memory cells and a single data line may be associated with more than one logical column. Memory cells (not shown in FIG. 1B) of at least a portion of array of memory cells **104** are capable of being programmed to one of at least two target data states.

[0041] Row decode circuitry **108** and column decode circuitry **109** are provided to decode address signals.

Address signals are received and decoded to access the array of memory cells 104. Memory device 130 also includes input/output (I/O) control circuitry 160 to manage input of commands, addresses and data to the memory device 130 as well as output of data and status information from the memory device 130. An address register 114 is in communication with I/O control circuitry 160 and row decode circuitry 108 and column decode circuitry 109 to latch the address signals prior to decoding. A command register 124 is in communication with I/O control circuitry 160 and local media controller 135 to latch incoming commands.

[0042] A controller (e.g., the local media controller 135 internal to the memory device 130) controls access to the array of memory cells 104 in response to the commands and generates status information for the external memory sub-system controller 115, i.e., the local media controller 135 is configured to perform access operations (e.g., read operations, programming operations and/or erase operations) on the array of memory cells 104. The local media controller 135 is in communication with row decode circuitry 108 and column decode circuitry 109 to control the row decode circuitry 108 and column decode circuitry 109 in response to the addresses. As described herein, local media controller 135 can perform source bias temperature compensation for read and program verify operations being performed on memory array 104. In one embodiment, local media controller 135 is in communication with a temperature sensor 180 disposed within or adjacent to memory device 130. Temperature sensor 180 can be used to measure an ambient temperature at certain points in time, which can represent, for example, a write temperature or a read temperature, and is otherwise referred to herein as an operating temperature.

[0043] The local media controller 135 is also in communication with a cache register 172. Cache register 172 latches data, either incoming or outgoing, as directed by the local media controller 135 to temporarily store data while the array of memory cells 104 is busy writing or reading, respectively, other data. During a program operation (e.g., write operation), data may be passed from the cache register 172 to the data register 170 for transfer to the array of memory cells 104; then new data may be latched in the cache register 172 from the I/O control circuitry 160. During a read operation, data may be passed from the cache register 172 to the I/O control circuitry 160 for output to the memory sub-system controller 115; then new data may be passed from the data register 170 to the cache register 172. The cache register 172 and/or the data register 170 may form (e.g., may form a portion of) a page buffer of the memory device 130. A page buffer may further include sensing devices (not shown in FIG. 1B) to sense a data state of a memory cell of the array of memory cells 104, e.g., by sensing a state of a data line connected to that memory cell. A status register 122 may be in communication with I/O control circuitry 160 and the local media controller 135 to latch the status information for output to the memory sub-system controller 115.

[0044] Memory device 130 receives control signals at the memory sub-system controller 115 from the local media controller 135 over a control link 132. For example, the control signals can include a chip enable signal CE #, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WE #, a read enable signal RE #, and a write protect signal WP #. Additional or alternative control signals (not shown) may be further

received over control link 132 depending upon the nature of the memory device 130. In one embodiment, memory device 130 receives command signals (which represent commands), address signals (which represent addresses), and data signals (which represent data) from the memory sub-system controller 115 over a multiplexed input/output (I/O) bus 236 and outputs data to the memory sub-system controller 115 over I/O bus 236.

[0045] For example, the commands may be received over input/output (I/O) pins [7:0] of I/O bus 236 at I/O control circuitry 160 and may then be written into command register 124. The addresses may be received over input/output (I/O) pins [7:0] of I/O bus 236 at I/O control circuitry 160 and may then be written into address register 114. The data may be received over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device at I/O control circuitry 160 and then may be written into cache register 172. The data may be subsequently written into data register 170 for programming the array of memory cells 104.

[0046] In an embodiment, cache register 172 may be omitted, and the data may be written directly into data register 170. Data may also be output over input/output (I/O) pins [7:0] for an 8-bit device or input/output (I/O) pins [15:0] for a 16-bit device. Although reference may be made to I/O pins, they may include any conductive node providing for electrical connection to the memory device 130 by an external device (e.g., the memory sub-system controller 115), such as conductive pads or conductive bumps as are commonly used.

[0047] It will be appreciated by those skilled in the art that additional circuitry and signals can be provided, and that the memory device 130 of FIG. 1B has been simplified. It should be recognized that the functionality of the various block components described with reference to FIG. 1B may not necessarily be segregated to distinct components or component portions of an integrated circuit device. For example, a single component or component portion of an integrated circuit device could be adapted to perform the functionality of more than one block component of FIG. 1B. Alternatively, one or more components or component portions of an integrated circuit device could be combined to perform the functionality of a single block component of FIG. 1B. Additionally, while specific I/O pins are described in accordance with popular conventions for receipt and output of the various signals, it is noted that other combinations or numbers of I/O pins (or other I/O node structures) may be used in the various embodiments.

[0048] FIG. 2 is a schematic of portions of an array of memory cells 104, such as a NAND memory array, as could be used in a memory of the type described with reference to FIG. 1B according to an embodiment. Memory array 104 includes access lines, such as wordlines 202₀ to 202_N, and data lines, such as bit lines 204₀ to 204_M. The wordlines 202 can be connected to global access lines (e.g., global wordlines), not shown in FIG. 2, in a many-to-one relationship. For some embodiments, memory array 104 can be formed over a semiconductor that, for example, can be conductively doped to have a conductivity type, such as a p-type conductivity, e.g., to form a p-well, or an n-type conductivity, e.g., to form an n-well.

[0049] Memory array 104 can be arranged in rows (each corresponding to a wordline 202) and columns (each corresponding to a bit line 204). Each column can include a string

of series-connected memory cells (e.g., non-volatile memory cells), such as one of NAND strings **206₀** to **206_M**. Each NAND string **206** can be connected (e.g., selectively connected) to a common source (SRC) **216** and can include memory cells **208₀** to **208_N**. The memory cells **208** can represent non-volatile memory cells for storage of data. The memory cells **208** of each NAND string **206** can be connected in series between a select gate **210** (e.g., a field-effect transistor), such as one of the select gates **210₀** to **210_M** (e.g., that can be source select transistors, commonly referred to as select gate source), and a select gate **212** (e.g., a field-effect transistor), such as one of the select gates **212₀** to **212_M** (e.g., that can be drain select transistors, commonly referred to as select gate drain). Select gates **210₀** to **210_M** can be commonly connected to a select line **214**, such as a source select line (SGS), and select gates **212₀** to **212_M** can be commonly connected to a select line **215**, such as a drain select line (SGD). Although depicted as traditional field-effect transistors, the select gates **210** and **212** can utilize a structure similar to (e.g., the same as) the memory cells **208**. The select gates **210** and **212** can represent a number of select gates connected in series, with each select gate in series configured to receive a same or independent control signal.

[0050] A source of each select gate **210** can be connected to common source **216**. The drain of each select gate **210** can be connected to a memory cell **208₀** of the corresponding NAND string **206**. For example, the drain of select gate **210₀** can be connected to memory cell **208₀** of the corresponding NAND string **206₀**. Therefore, each select gate **210** can be configured to selectively connect a corresponding NAND string **206** to the common source **216**. A control gate of each select gate **210** can be connected to the select line **214**.

[0051] The drain of each select gate **212** can be connected to the bit line **204** for the corresponding NAND string **206**. For example, the drain of select gate **212₀** can be connected to the bit line **204₀** for the corresponding NAND string **206₀**. The source of each select gate **212** can be connected to a memory cell **208_N** of the corresponding NAND string **206**. For example, the source of select gate **212₀** can be connected to memory cell **208_N** of the corresponding NAND string **206₀**. Therefore, each select gate **212** can be configured to selectively connect a corresponding NAND string **206** to the corresponding bit line **204**. A control gate of each select gate **212** can be connected to select line **215**.

[0052] The memory array **104** in FIG. 2 can be a quasi-two-dimensional memory array and can have a generally planar structure, e.g., where the common source **216**, NAND strings **206** and bit lines **204** extend in substantially parallel planes. Alternatively, the memory array **104** in FIG. 2 can be a three-dimensional memory array, e.g., where NAND strings **206** can extend substantially perpendicular to a plane containing the common source **216** and to a plane containing the bit lines **204** that can be substantially parallel to the plane containing the common source **216**.

[0053] Typical construction of memory cells **208** includes a data-storage structure **234** (e.g., a floating gate, charge trap, and the like) that can determine a data state of the memory cell (e.g., through changes in threshold voltage), and a control gate **236**, as shown in FIG. 2. The data-storage structure **234** can include both conductive and dielectric structures while the control gate **236** is generally formed of one or more conductive materials. In some cases, memory cells **208** can further have a defined source/drain (e.g., source) **230** and a defined source/drain (e.g., drain) **232**. The

memory cells **208** have their control gates **236** connected to (and in some cases form) a wordline **202**.

[0054] A column of the memory cells **208** can be a NAND string **206** or a number of NAND strings **206** selectively connected to a given bit line **204**. A row of the memory cells **208** can be memory cells **208** commonly connected to a given wordline **202**. A row of memory cells **208** can, but need not, include all the memory cells **208** commonly connected to a given wordline **202**. Rows of the memory cells **208** can often be divided into one or more groups of physical pages of memory cells **208**, and physical pages of the memory cells **208** often include every other memory cell **208** commonly connected to a given wordline **202**. For example, the memory cells **208** commonly connected to wordline **202_N** and selectively connected to even bit lines **204** (e.g., bit lines **204₀**, **204₂**, **204₄**, etc.) can be one physical page of the memory cells **208** (e.g., even memory cells) while memory cells **208** commonly connected to wordline **202_N** and selectively connected to odd bit lines **204** (e.g., bit lines **204₁**, **204₃**, **204₅**, etc.) can be another physical page of the memory cells **208** (e.g., odd memory cells).

[0055] Although bit lines **204₃**-**204₅** are not explicitly depicted in FIG. 2, it is apparent from the figure that the bit lines **204** of the array of memory cells **104** can be numbered consecutively from bit line **204₀** to bit line **204_M**. Other groupings of the memory cells **208** commonly connected to a given wordline **202** can also define a physical page of memory cells **208**. For certain memory devices, all memory cells commonly connected to a given wordline can be deemed a physical page of memory cells. The portion of a physical page of memory cells (which, in some embodiments, could still be the entire row) that is read during a single read operation or programmed during a single programming operation (e.g., an upper or lower page of memory cells) can be deemed a logical page of memory cells. A block of memory cells can include those memory cells that are configured to be erased together, such as all memory cells connected to wordlines **202₀**-**202_N** (e.g., all NAND strings **206** sharing common wordlines **202**). Unless expressly distinguished, a reference to a page of memory cells herein refers to the memory cells of a logical page of memory cells. Although the example of FIG. 2 is discussed in conjunction with NAND flash, the embodiments and concepts described herein are not limited to a particular array architecture or structure, and can include other structures (e.g., SONOS, phase change, ferroelectric, etc.) and other architectures (e.g., AND arrays, NOR arrays, etc.).

[0056] FIG. 3A is a schematic of portions of an array of memory cells implementing source bias temperature compensation for read and program verify operations in accordance with some embodiments of the present disclosure. The portion of the array of memory cells, such as memory array **104**, can be a block **300**, for example. In one embodiment, the block **300** includes strings of memory cells that can be grouped into sub-blocks, such as sub-blocks **305₀**-**305₃**. Other numbers of sub-blocks can be included in other embodiments.

[0057] Specifically, in at least some embodiments, the block **300** includes a bit line **304**, where each sub-block is coupled to the bit line **304**. The first sub-block **305₀** can include a first drain select (SGD) transistor **312₀**, a first source select (SGS) transistor **310₀**, and a first string of memory cells **306₀** coupled therebetween. The second sub-block **305₁** can include a second SGD transistor **312₁**, a

second SGS transistor **310**₁, and a second string of memory cells **306**₁ coupled therebetween. The third sub-block **305**₂ can include a third SGD transistor **312**₂, a third SGS transistor **310**₂, and a third string of memory cells **306**₂ coupled therebetween. The fourth sub-block **305**₃ can include a fourth SGD transistor **312**₃, a fourth SGS transistor **310**₃, and a fourth string of memory cells **306**₃ coupled therebetween. By way of example, the first string of memory cells **306**₀ includes multiple memory cells **308**₀ . . . **308**_N. Each SGS transistor can be connected to a common source (SRC), such as a source voltage line **303**, to provide voltage to the sources of the multiple memory cells **308**₀ . . . **308**_N. In some embodiments, the source voltage line **303** includes a source plate that supplies the source voltage (e.g., in the form of a source voltage signal). In at least some embodiments, multiple wordlines (WLs) are coupled with gates of memory cells of each string of memory cells **306**₀ . . . **306**₃.

[0058] In these embodiments, a first drain select gate line (SGD0) can be connected to the gate of the first SGD transistor **312**₀, a second drain select gate line (SGD1) can be connected to the gate of the second SGD transistor **312**₁, a third drain select gate line (SGD2) can be connected to the gate of the third SGD transistor **312**₂, and a fourth drain select gate line (SGD3) can be connected to the gate of the fourth SGD transistor **312**₃. Further, a first source select gate line (SGS0) can be connected to the gate of the first SGS transistor **310**₀, a second source select gate line (SGS1) can be connected to the gate of the second SGS transistor **310**₁, a third source select gate line (SGS2) can be connected to the gate of the third SGS transistor **310**₂, and a fourth source select gate line (SGS3) can be connected to the gate of the fourth SGS transistor **310**₃.

[0059] In one embodiment, local media controller **135** can perform source bias temperature compensation for read and program verify operations being performed on block **300**. In one embodiment, the magnitude of a source voltage signal applied to source voltage line **303** can be adjusted based on a temperature associated with the memory device at the time the read or program verify operation is performed. For example, local media controller **135** can determine a temperature compensation value associated with a measured operating temperature and modify a default magnitude of the source voltage signal by an amount corresponding to the temperature compensation value. As illustrated in FIG. 3B, in general, as the measured operating temperature increases, the magnitude of the source voltage signal (e.g., source bias **452**) applied to source voltage line **303** also increases. During such a memory access operation, the bitline voltage signal (e.g., bitline bias **450**) applied to bitline **304** remains at a constant magnitude, even as the operating temperature changes. To perform the memory access operation, local media controller **135** can cause the modified source voltage signal to be applied to source voltage line **303** and cause at least one of a read voltage signal or a program verify voltage signal to be applied to the selected wordline WLx. The temperature compensated source voltage on source voltage line **303** increases the drain-to-source voltage differential (V_{ds}) of a selected memory cell **308**_x, which, in turn, causes an increased string current through the selected memory string **306**₀.

[0060] FIG. 4 is a flow diagram of an example method of source bias temperature compensation for read and program verify operations on a memory device of a memory sub-system in accordance with some embodiments of the present

disclosure. The method **400** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **400** is performed by local media controller **135** of FIG. 1A and FIG. 1B. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

[0061] At operation **405**, a request is received. For example, control logic (e.g., local media controller **135**) can receive a request to perform a memory access operation on a memory array, such as memory array **104**, of a memory device, such as memory device **130**. In one embodiment, the request is received from a requestor, such as memory interface **113** of memory sub-system controller **115**, or host system **120**. In one embodiment, the request to perform the memory access operation includes a request to perform a read operation, such as to read data store at one or more memory cells of memory array **104**, or a request to perform a program verify operation, such as to verify that data was successfully programmed to one or more memory cells of memory array **104**.

[0062] At operation **410**, an operating temperature is determined. For example, the control logic can determine the operating temperature at a time when the request to perform the memory operation is received. In one embodiment, the control logic can obtain a temperature measurement value from a temperature sensor on the memory device **130**, such as temperature sensor **180** (i.e., an on-die temperature sensor). Depending on the embodiment, the control logic can either query temperature sensor **180** for a new temperature measurement in response to receiving the request at operation **405**, or can use a most recently measured temperature value (e.g., when temperature measurements are routinely taken at periodic intervals on memory device **130**). In another embodiment, the control logic can receive a temperature measurement value from a temperature sensor external to the memory device, such as a sensor located elsewhere in memory sub-system **110**.

[0063] At operation **415**, a voltage is modified. For example, the control logic can modify a default magnitude of a source voltage signal, such as source bias **452**, based on the operating temperature to form a modified source voltage signal. In one embodiment, local media controller **135** determines a temperature compensation value associated with the operating temperature. For example, local media controller **135** can calculate an intermediate temperature compensation value using a defined formula. The measured operating temperature can be an input to the formula, and the intermediate temperature compensation value can be an output. Local media controller **135** can further identify the temperature compensation value based on a granularity of a source voltage signal driver. For example, the source voltage signal driver may be regulated to provide digital voltage levels in certain increments (e.g., 10 millivolt increments). If

the calculated intermediate temperature compensation value does not correspond to one of those increments, local media controller **135** can identify a closest (or next highest) increment, and use that value as the temperature compensation value. In another embodiment, local media controller **135** can identify a corresponding entry of a plurality of entries in a data structure, such as a look-up table, stored on memory device **130**. The corresponding entry can be associated with the identified operating temperature and can include an indication of the temperature compensation value (e.g., which was predetermined).

[0064] Upon determining the temperature compensation value, the control logic can modify the default magnitude of the source voltage signal by an amount corresponding to the temperature compensation value. In one embodiment, the source voltage signal can have a default magnitude that is used in the absence of any temperature compensation. The default magnitude can be set based on a default temperature (e.g., 0 degrees Celsius, a normal operating temperature, or some other temperature). In one embodiment, local media controller **135** can increase (or decrease) the default magnitude by an amount corresponding to the temperature compensation value to generate a modified or adjusted magnitude. As indicated in FIG. 3B, the adjusted magnitude of source bias **452** at a higher temperature will typically be larger than the adjusted magnitude (or default magnitude) of source bias **452** at a lower temperature.

[0065] At operation **420**, a signal is applied. For example, the control logic can cause the modified source voltage signal to be applied to the memory array **104**. In one embodiment, local media controller **135** directs a signal driver to apply the modified source voltage signal to source signal line **303** of block **300** of the memory array **104**. As described above, the modified source voltage signal can have a magnitude adjusted in view of the measured operating temperature of the memory device **130**.

[0066] At operation **425**, a memory access operation is performed. For example, the control logic can perform the memory access operation, such as a read operation or a program verify operation, on the memory array **104**. In one embodiment, local media controller **135** directs a signal driver to apply at least one of a read voltage signal or a program verify voltage signal to a selected wordline, such as wordline WLx, associated with a selected memory cell, such as memory cell **308x**. In one embodiment, the read voltage signal or program verify voltage signal is applied to the selected wordline concurrently (e.g., at least partially overlapping in time) with the modified source voltage signal being applied to the source signal line **303**. As described above, the modified source voltage signal on source voltage line **303** increases the drain-to-source voltage differential (Vds) of the selected memory cell **308x** and an increased string current through the selected memory string **306₀**, which limits threshold voltage distribution widening and allows the read or program verify operations to be performed with a larger read window budget and fewer errors.

[0067] FIG. 5 is a diagram illustrating example waveforms for source bias temperature compensation for read and program verify operations on a memory device in accordance with some embodiments of the present disclosure. Three different example operating temperatures are illustrated, including a low temperature, a medium temperature (which is higher than the low temperature), and a high temperature (which is higher than the medium temperature).

At each operating temperature, the signals applied to the selected wordline can be the same, including a first pulse **562** (e.g., a pass voltage applied to all unselected wordlines, as well as the selected wordline) to open the channels of the respective memory strings in the memory array, and a series of one or more pulses **564** (e.g., a read voltage or program verify voltage) that incrementally increase as the memory access operation is performed on memory cells programmed to different levels. In other embodiments, the signals applied to the selected wordline can vary with temperature. For example, the control logic can utilize a program verify/read temperature compensation value to adjust the voltage applied to the selected wordline to improve the median threshold voltage shift alignment. As illustrated in FIG. 5, however, the magnitude of the source bias signal applied to the source signal line **303** varies at each of the operating temperature. At the low temperature, the source bias signal has a low magnitude **572**, at the medium temperature, the source bias signal has a medium magnitude **574** (which is higher than the low magnitude **572**), and at the high temperature, the source bias signal has a high magnitude **576** (which is higher than the medium magnitude **574**).

[0068] FIG. 6 illustrates an example machine of a computer system **600** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In some embodiments, the computer system **600** can correspond to a host system (e.g., the host system **120** of FIG. 1) that includes, is coupled to, or utilizes a memory sub-system (e.g., the memory sub-system **110** of FIG. 1) or can be used to perform the operations of a controller (e.g., to execute an operating system to perform operations corresponding to the local media controller **135** of FIG. 1). In alternative embodiments, the machine can be connected (e.g., networked) to other machines in a LAN, an intranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

[0069] The machine can be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0070] The example computer system **600** includes a processing device **602**, a main memory **604** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **606** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage system **618**, which communicate with each other via a bus **630**.

[0071] Processing device **602** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device can be a complex instruction set computing (CISC) microprocessor, reduced instruction set

computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device 602 can also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device 602 is configured to execute instructions 626 for performing the operations and steps discussed herein. The computer system 600 can further include a network interface device 608 to communicate over the network 620.

[0072] The data storage system 618 can include a machine-readable storage medium 624 (also known as a computer-readable medium) on which is stored one or more sets of instructions 626 or software embodying any one or more of the methodologies or functions described herein. The instructions 626 can also reside, completely or at least partially, within the main memory 604 and/or within the processing device 602 during execution thereof by the computer system 600, the main memory 604 and the processing device 602 also constituting machine-readable storage media. The machine-readable storage medium 624, data storage system 618, and/or main memory 604 can correspond to the memory sub-system 110 of FIG. 1.

[0073] In one embodiment, the instructions 626 include instructions to implement functionality corresponding to the local media controller 135 of FIG. 1). While the machine-readable storage medium 624 is shown in an example embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media, and magnetic media.

[0074] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0075] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. The present disclosure can refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into

other data similarly represented as physical quantities within the computer system memories or registers or other such information storage systems.

[0076] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus can be specially constructed for the intended purposes, or it can include a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program can be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0077] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems can be used with programs in accordance with the teachings herein, or it can prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages can be used to implement the teachings of the disclosure as described herein.

[0078] The present disclosure can be provided as a computer program product, or software, that can include a machine-readable medium having stored thereon instructions, which can be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). In some embodiments, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory components, etc.

[0079] In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications can be made thereto without departing from the broader spirit and scope of embodiments of the disclosure as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A memory device comprising:

a memory array comprising a plurality of strings of memory cells coupled between a bitline and a source terminal, wherein respective memory cells in each of the plurality of strings of memory cells are coupled to respective wordlines; and

- control logic, operatively coupled with the memory array, to perform operations comprising:
- causing a bitline voltage signal with a fixed magnitude to be applied to the bitline of the memory array during a memory access operation;
 - causing a source voltage signal having a temperature compensated magnitude to be applied to the source terminal of the memory array during the memory access operation; and
 - causing a memory access operation voltage signal to be applied to one or more wordlines of the memory array during the memory access operation.
2. The memory device of claim 1, wherein the memory access operation comprises a read operation.
 3. The memory device of claim 1, wherein the memory access operation comprises a program-verify operation.
 4. The memory device of claim 1, further comprising: an on-die temperature sensor, wherein the control logic is to perform operations further comprising:
 - determining an operating temperature of the memory device by obtaining a temperature measurement from the on-die temperature sensor.
 5. The memory device of claim 4, wherein the temperature compensated magnitude of the source voltage signal is based on the operating temperature of the memory device.
 6. The memory device of claim 5, where the control logic is to perform operations further comprising:
 - determining the temperature compensated magnitude of the source voltage signal by:
 - determining a temperature compensation value associated with the operating temperature; and
 - modifying a default magnitude of the source voltage signal by an amount corresponding to the temperature compensation value.
 7. The memory device of claim 6, wherein determining the temperature compensation value associated with the operating temperature comprises calculating an intermediate temperature compensation value and identifying the temperature compensation value from the intermediate temperature compensation value based on a granularity of a source voltage signal driver.
 8. The memory device of claim 6, wherein determining the temperature compensation value associated with the operating temperature comprises identifying a corresponding entry of a plurality of entries in a look-up table, wherein the corresponding entry is associated with the operating temperature and comprises an indication of the temperature compensation value.
 9. A method comprising:
 - causing a bitline voltage signal with a fixed magnitude to be applied to a bitline of a memory array of a memory device during a memory access operation, the memory array comprising a plurality of strings of memory cells coupled between a bitline and a source terminal, wherein respective memory cells in each of the plurality of strings of memory cells are coupled to respective wordlines;
 - causing a source voltage signal having a temperature compensated magnitude to be applied to the source terminal of the memory array during the memory access operation; and
 - causing a memory access operation voltage signal to be applied to one or more wordlines of the memory array during the memory access operation.
 10. The method of claim 9, wherein the memory access operation comprises a read operation.
 11. The method of claim 9, wherein the memory access operation comprises a program-verify operation.
 12. The method of claim 9, wherein the memory device further comprises an on-die temperature sensor, the method further comprising:
 - determining an operating temperature of the memory device by obtaining a temperature measurement from the on-die temperature sensor.
 13. The method of claim 12, wherein the temperature compensated magnitude of the source voltage signal is based on the operating temperature of the memory device.
 14. The method of claim 13, further comprising:
 - determining the temperature compensated magnitude of the source voltage signal by:
 - determining a temperature compensation value associated with the operating temperature; and
 - modifying a default magnitude of the source voltage signal by an amount corresponding to the temperature compensation value.
 15. The method of claim 14, wherein determining the temperature compensation value associated with the operating temperature comprises calculating an intermediate temperature compensation value and identifying the temperature compensation value from the intermediate temperature compensation value based on a granularity of a source voltage signal driver.
 16. The method of claim 14, wherein determining the temperature compensation value associated with the operating temperature comprises identifying a corresponding entry of a plurality of entries in a look-up table, wherein the corresponding entry is associated with the operating temperature and comprises an indication of the temperature compensation value.
 17. A memory device comprising:
 - a memory array comprising a plurality of strings of memory cells coupled between a bitline and a source terminal, wherein respective memory cells in each of the plurality of strings of memory cells are coupled to respective wordlines; and
 - control logic, operatively coupled with the memory array, to perform operations comprising:
 - causing a bitline voltage signal with a fixed magnitude to be applied to the bitline of the memory array during a memory access operation;
 - generating a source voltage signal having a temperature compensated magnitude; and
 - performing at least one of a read operation or a program verify operation on the memory array when the source voltage signal having the temperature compensated magnitude is applied to the source terminal of the memory array.
 18. The memory device of claim 17, further comprising: an on-die temperature sensor, wherein the control logic is to perform operations further comprising:
 - determining an operating temperature of the memory device by obtaining a temperature measurement from the on-die temperature sensor.
 19. The memory device of claim 18, wherein the temperature compensated magnitude of the source voltage signal is based on the operating temperature of the memory device.

20. The memory device of claim **19**, wherein generating the source voltage signal having the temperature compensated magnitude comprises:

determining a temperature compensation value associated with the operating temperature; and
modifying a default magnitude of the source voltage signal by an amount corresponding to the temperature compensation value.

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